

Breast Cancer Detection Using Deep Learning and Quantum-Inspired Hyperparameter Optimization

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Abstract—Breast cancer remains one of the most prevalent causes of mortality among women that highlights the critical need for accurate and early detection. The objective of this study is to compare the deep learning-based computer-aided diagnosis (CAD) systems for breast cancer detection, to explore the impact of hyperparameter optimization techniques, and to propose an ensemble model with optimized hyperparameter to improve classification accuracy for breast cancer detection. To attain these objectives, a representative subset of mammographic data was used to assess the performance of state-of-the-art pretrained models, including VGG19, ResNet50, MobileNet, InceptionV3, and DenseNet. It has been observed that although the classical tuning methods (e.g., Bayesian optimization) significantly enhanced model accuracy, but incurred high computational costs. Thus, a quantum-inspired optimization strategy based on the Quadratic Unconstrained Binary Optimization (QUBO) formulation was employed here. Despite the reduced data volume and computational load, the QUBO-based approach achieved competitive diagnostic performance. Furthermore, an ensemble model with optimized hyperparameter was introduced to mitigate overfitting and improve generalizability for yielding an accuracy of 92% with balanced precisions and recalls. The findings firstly underscore the complementary strengths of classical and quantum-inspired tuning, and secondly, their combined application within ensemble frameworks that offer a robust and scalable pathway for enhancing CAD systems in breast cancer screening.

Index Terms—Breast Cancer Detection, Deep Learning, QUBO Optimization, Ensemble Learning, Hyperparameter Tuning, CBIS-DDSM, Medical Imaging

I. INTRODUCTION

Breast cancer remains a significant global health issue, particularly in women, and is associated with high mortality rates. Although mammography is the most widely used tool for early detection, distinguishing between benign and malignant tumors is often challenging owing to the overlapping visual features of lesions [1], [2]. Currently, CAD systems enhanced by deep learning are proving to be accurate in diagnosing breast cancer. For example, studies leveraging the CBIS-DDSM dataset have shown promising classification results through deep learning [3]. However, a high-dimensional large dataset requires out-of-capacity memory space and a highly configured CPU for model training, tuning, and ensembling. Moreover, overfitting owing to the complexity of deep learning models remains an issue in detecting breast cancer [4].

CAD systems have revolutionized breast cancer detection, leading to faster and more accurate diagnosis. Convolutional Neural Networks (CNNs) are particularly well-suited for analyzing mammographic images, which helps detect and classify tumors with high precision. Similarly, pre-trained models such as VGG16, ResNet, MobileNet, InceptionV3, and DenseNet enhance the classification accuracy by adding a customized layer and hyperparameter tuning [5]. Moreover,

ensemble methods that combine multiple models can significantly reduce overfitting issues during model validation [6].

Optimization of these models requires fine-tuning of hyperparameters, such as the learning rate and batch size. Traditional methods, such as Grid Search and Bayesian Optimization, are commonly used for this purpose and are comparatively computationally expensive and time-consuming [7]. In other studies, the authors explored methods such as Simulated Annealing and Bayesian Optimization to improve efficiency and model performance [8].

Quantum-inspired annealing, often emulated via simulated annealing using QUBO formulations, has shown promise in hyperparameter tuning and feature selection for deep learning. While underexplored in breast cancer detection, comparing classical and quantum-inspired methods can provide insights into optimizing the classification performance [9].

This work makes the following original contributions to breast cancer CAD: (i) Benchmarking with classical tuning to analyze the performance of deep learning models using the traditional hyperparameter tuning approach for detecting breast cancer (ii) Quantum-inspired hyperparameter optimization to analyze the performance of detecting breast cancer (iii) Ensemble model with optimized hyperparameters to improve classification robustness, accuracy, and stable generalization for breast cancer detection.

The remainder of this paper is organized as follows. Section II provides a review of related literature. Section III describes the proposed methodology. Section IV discusses experimental results. Finally, Section V presents a detailed discussion and conclusions.

II. LITERATURE REVIEW

Deep learning has significantly advanced breast cancer detection in medical imaging, particularly with CNNs, achieving high mammogram analysis accuracy. Early work by Ribli et al. [10], demonstrated that an end-to-end CNN (using a Faster R-CNN framework) could automatically detect and classify breast lesions on mammograms without manual intervention, marking a leap over traditional CAD methods. Subsequent studies explored hybrid classification approaches. For example, Ragab et al. [11], combined CNN-based feature extraction with a support vector machine (SVM) classifier to improve malignant/benign classification performance on the Digital Database for Screening Mammography (DDSM). As public data became available, the Curated Breast Imaging Subset of DDSM (CBIS-DDSM) has emerged as a standard benchmark for deep learning models. Tsochatzidis et al. [12], conducted a comprehensive comparison of architectures (Inception-v2, ResNet-50/101/152, VGG, etc.) fine-tuned on CBIS-DDSM, finding that transfer learning with deeper CNNs yields robust accuracy in lesion classification. Similarly, Arora et al. [13], adopted pretrained deep features in an ensemble classifier to automatically distinguish malignant from benign mammograms. Ensemble and multi-CNN strategies have further improved performance. Ragab et al. [14], achieved a higher accuracy by fusing predictions from multiple deep models. Recent studies have reported record performance on the CBIS-DDSM

with improved data preprocessing and optimization. Jabeen et al. [15], achieved a high classification accuracy by augmenting images, fine-tuning an EfficientNet-based model, and then applying a custom feature selection metaheuristic.

Researchers have explored advanced hyperparameter optimization techniques to enhance the model performance. Liao et al. [16], improved reproducibility on CBIS-DDSM with an open-source pipeline and found that higher image resolution boosted CNN detection. Moving beyond grid search, metaheuristics such as Simulated Annealing (SA) have shown promise. Bahrami et al. [17], demonstrated that SA-optimized ensembles achieved higher sensitivity than both untuned and AutoML-based models. SA, a classical proxy for quantum annealing (QA) using a QUBO formulation, is supported by recent research showing that it can effectively replicate the key behaviors of QA under certain conditions. For example, Mbeng et al. [18], demonstrated closely mimics QA's scaling properties of quantum-inspired annealing in ordered systems. Again, Wauters et al. [19], highlighted its utility for performance analysis, where direct access to quantum hardware is limited.

Although most existing models rely on static feature extractors, few have simultaneously addressed the dual challenges of feature redundancy and efficient hyperparameter space exploration. Many prior studies lack robust regularization strategies, often resulting in overfitting, particularly during the validation phases. Moreover, traditional tuning methods such as Grid Search or Bayesian Optimization are computationally expensive and unsuitable for high-dimensional parameter spaces, whereas quantum-inspired or metaheuristic approaches face scalability and convergence limitations. In light of these constraints, our study introduces an ensemble of deep learning models integrated with both traditional and quantum-inspired annealing-based hyperparameter optimization. By employing early stopping and dropout regularization during model training, we mitigated overfitting and enhanced generalization. This approach not only improves classification accuracy but also reduces training overhead, presenting a viable solution for real-world, scalable, and intelligent CAD systems for breast cancer detection.

III. RESEARCH METHODOLOGY

A. Dataset Preprocessing

We evaluated our approach on the Breast Imaging Subset of DDSM (CBIS-DDSM), a public mammography dataset containing 10,239 scanned film mammograms (approximately 6 GB of JPEG images). Each image was annotated with a region of interest (ROI) delineating suspicious lesions. For our classification task, we consolidated the dataset labels into two categories: malignant (cancerous) and benign (no cancerous), which yields a binary classification problem. Before model training, we performed thorough data preprocessing and augmentation steps to enhance the image quality and model generalization. Figure 1 provides an overview of the processing pipeline file.

To focus the models on the tumor regions, we utilized the provided ROI masks and generated binary ROI masks from the mammograms using intensity thresholding by comparing the pixel intensities to a threshold value T (see Figure 2). The mask M is a binary image where pixels corresponding to the cancerous region were marked as 1 (or 255), and the rest of the pixels are 0. After generating the binary mask, to extract only the region of interest (ROI) that contains cancer cells. This is done by performing an element-wise multiplication between the image I and the mask M (see equation i):

$$I_{\text{ROI}} = I \cdot M \quad (\text{i})$$

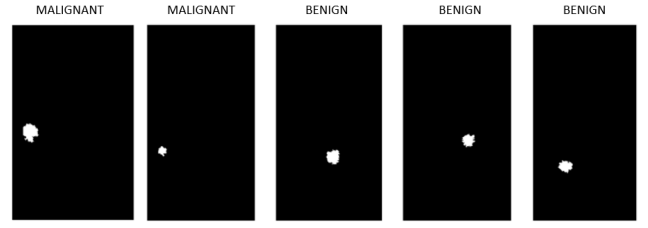


Fig. 2: ROI Mask Generation from Input Mammogram

After ROI masking, we applied standard image augmentations (random rotations, flips, scaling, etc.) to artificially enlarge the training set and reduce overfitting. All images were then resized to a uniform resolution (224 x 224 pixels) appropriate for deep networks, and pixel intensity normalization was performed as required by each model's preprocessing pipeline.

The dataset was split into three sets: 7,167 images were designated for training, 2,048 images for validation, and 1,024 images for testing. This split ensures that the model has adequate data for training while reserving a separate subset for validation and testing to evaluate its performance. For quantum-inspired annealing tuning on the individual models and ensemble deep learning, 10% sample data was chosen to feed the model for training due to limited memory availability.

B. Deep Learning Models and Ensemble Strategy

We experimented with several deep learning models that are representative of state-of-the-art CNN architectures for medical imaging [20]. In particular, we trained and evaluated the following six models on the preprocessed CBIS-DDSM dataset:

(a) Custom CNN, a baseline CNN designed for mammography classification (with multiple convolutional and pooling layers, followed by fully connected layers). (b) ResNet50, a 50-layer residual network known for its deep architecture and skip connections, and is used here via transfer learning for feature extraction. (c) VGG16, a 16-layer CNN with a sequential architecture of conv/pool layers, was included for its proven performance in image recognition tasks. (d) MobileNet, a lightweight CNN architecture optimized for efficiency to test if a smaller model can perform adequately on mammograms. (e) DenseNet121, a 121-layer densely connected network that reuses features through dense blocks, is potentially beneficial for capturing subtle radiographic features. (f) InceptionV3, a 48-layer CNN with inception modules that capture multiscale features, which can be useful for complex textures in mammograms.

All deep networks (except the custom CNN) were initialized with ImageNet pretrained weights and fine-tuned on our mammography data to leverage transfer learning. Each model outputs a probability for the malignant class, which is thresholded to produce a final binary prediction (malignant or benign).

In addition to the individual models, we employed an ensemble learning approach to further improve robustness. Specifically, we combined three complementary models – custom CNN, ResNet101, and DenseNet121 into an ensemble classifier. The ensemble aggregated the predictions of its members for each input image. In our implementation, we used a simple soft-voting scheme: the predicted class probabilities from the three models were averaged, and the average was used to determine the final class label. We note that all constituent models in the ensemble are trained on the same training set (with the limited dataset size for certain tuning methods, as described next), and their diversity comes from differences in architecture and optimization hyperparameters.

C. Hyperparameter Optimization Techniques

1) Traditional Hyperparameter Tuning: We applied several established search techniques to tune the models, including Grid

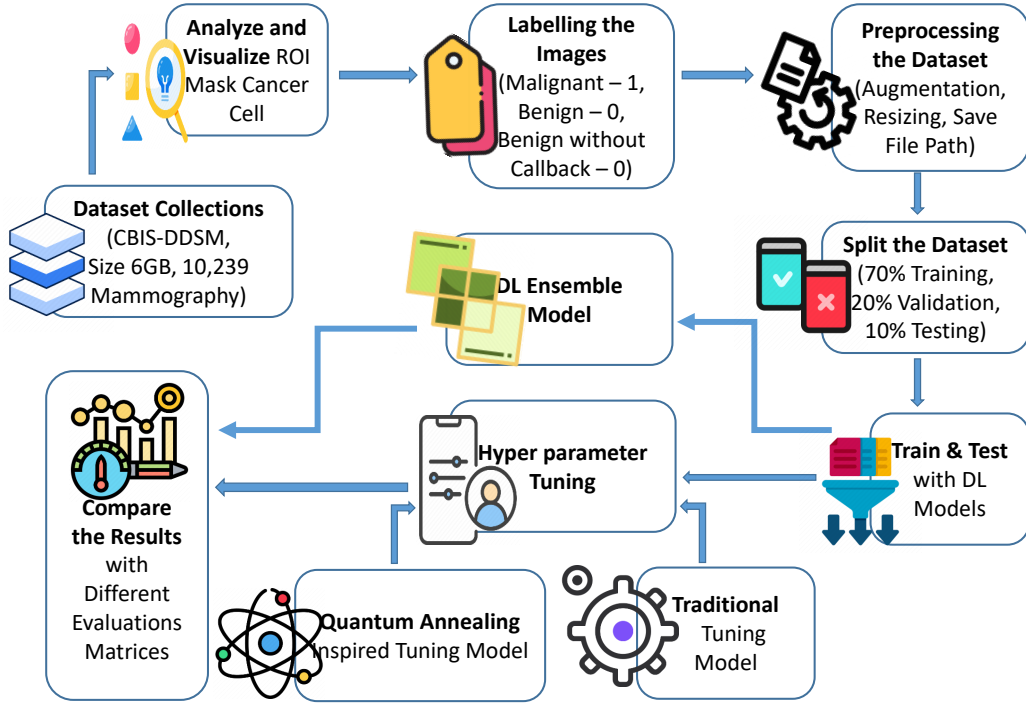


Fig. 1: Overview of the proposed methodology

Search, Bayesian Optimization, Hyperband, and Random Search. These methods (summarized in Algorithm 1) iteratively explore the predefined hyperparameter space and evaluate the model performance on the validation set to identify high-performing combinations. In our experiments, we defined a reasonable search range for each hyperparameter (learning rate, dropout ratio, etc.) based on prior studies and domain knowledge. Each model (CNN, ResNet50, VGG16, MobileNet, DenseNet121, and InceptionV3) was first trained using default or baseline hyperparameters to obtain a reference performance. We then performed tuning using the above algorithms to improve upon these baselines. The search process was constrained by computational resources, but the key hyperparameters were successfully optimized for all models. For each network, the output of this stage is a set of the best hyperparameters (denoted θ^*) that yielded the highest validation accuracy during the search. These optimal values were later used to retrain each model from scratch on the full training set (or subset, as noted below) before the final evaluation.

2) Quantum Annealing-inspired Tuning: We investigated an alternative optimization approach inspired by quantum annealing metaheuristics. It is an emerging technique that solves optimization problems by gradually evolving a quantum system towards a state that minimizes a given objective (often formulated as a QUBO problem). Directly applying quantum annealing to deep-learning hyperparameter tuning is challenging owing to the large combinatorial search space and limited quantum hardware. In our study, we emulate the quantum annealing process via a simulated annealing algorithm that stochastically searches for better hyperparameters. Algorithm 2 outlines the quantum-inspired simulated annealing procedure. Starting from an initial hyperparameter set, the algorithm randomly perturbs the hyperparameters to sample a new configuration, evaluates it on the validation set, and decides whether to accept it based on the change in performance. Worse configurations can occasionally be accepted according to the Boltzmann probability $P = \exp(-\Delta E/T)$ (where ΔE is the increase in the validation loss and T is a temperature parameter). As the algorithm iterates, T is gradually lowered, reducing the chance of accepting inferior solutions and thereby annealing

Algorithm 1 Traditional Hyperparameter Tuning Algorithm (Bayesian, Hyperband, Grid Search)

- 1: **Input:** Training data D , validation data D_{val} , predefined hyperparameter search space θ
 - 2: **Output:** Best hyperparameter configuration θ^*
 - 3: Initialize hyperparameter search space $\theta = \{\text{learningrate, denselayers, dropoutrate, activationfunction}\}$
 - 4: **for each tuning technique in** $\{\text{Bayesian, Hyperband, Grid Search}\}$:
 - 5: **for each combination of hyperparameters in search space:**
 - 6: Train model on D using hyperparameters θ
 - 7: Evaluate model on D_{val}
 - 8: Compute performance metrics $\mathcal{L}_{val}$ (e.g., accuracy, F1-score)
 - 9: **if** $\mathcal{L}_{val}$ improves:
 - 10: Set current hyperparameter combination as best θ^*
 - 11: **Return** θ^* (Best hyperparameters)
-

the system to a hopefully global-optimal hyperparameter set. This approach allows the escape of local optima early on and fine-tuning around a promising region of the hyperparameter space as the search progresses.

After obtaining the optimal hyperparameters from both the traditional and quantum-inspired approaches, we trained the final model instances for each approach. For large-scale traditional tuning, the models were trained on the full 7,167-image training set, with their optimized hyperparameters. For the quantum-inspired tuning, the selected models (CNN, ResNet101, and DenseNet121) were trained on the 1,000-sample dataset with their respective optimized

Algorithm 2 Quantum Annealing-Inspired Simulated Annealing Tuning

- 1: **Input:** Training data D , validation data D_{val} , initial temperature T_0 , hyperparameter search space θ
 - 2: **Output:** Best hyperparameter configuration θ^*
 - 3: Initialize temperature T_0 (initial exploration phase)
 - 4: Initialize best hyperparameter configuration $\theta^* = \emptyset$
 - 5: **while** temperature T is greater than threshold:
 - 6: Randomly sample new hyperparameter configuration θ'
 - 7: Train model on D with θ'
 - 8: Evaluate model on D_{val} to get performance $\mathcal{L}_{val}$
 - 9: Compute energy difference $E = \mathcal{L}_{val}(\theta') - \mathcal{L}_{val}(\theta^*)$
 - 10: **if** $E < 0$ (better configuration found):
 - 11: Update best configuration: $\theta^* = \theta'$
 - 12: **else if** random acceptance criteria met (based on temperature T):
 - 13: Accept new configuration with probability $P = \exp\left(\frac{-E}{T}\right)$
 - 14: Decrease temperature $T = \alpha \cdot T$ (annealing step)
 - 15: **Return** θ^*
-

hyperparameters. We also trained an ensemble model by combining the predictions of the three quantum-tuned models, as previously described. All the models were trained until convergence or until the validation loss stopped improving (early stopping), and the epoch with the highest validation accuracy was recorded for each. The next section details the performance results and comparisons between these models and the tuning strategies.

IV. RESULT ANALYSIS

This section comprehensively analyzes the experimental outcomes of applying both traditional and quantum annealing-inspired hyperparameter optimization techniques to various deep learning architectures for breast cancer detection presented in Figure 3.

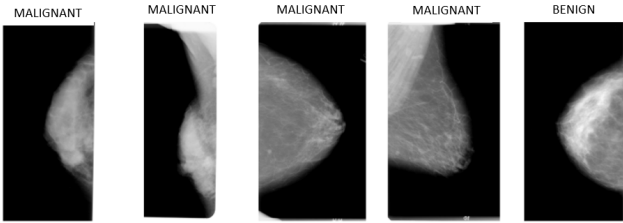


Fig. 3: Image Classification Result Sample

A. Evaluation Metrics and Experimental Setup

We evaluated model performance using a range of standard metrics to ensure a comprehensive analysis. These include accuracy (the proportion of correctly classified mammograms), precision (positive predictive value), recall (sensitivity or true positive rate), and F1-score (harmonic mean of precision and recall). We monitored the training and validation accuracy and loss curves during training to assess the convergence and detect overfitting. We also recorded the epoch with the best validation accuracy for each model to determine the training efficiency.

All the models were implemented in TensorFlow/Keras and trained on a workstation with an NVIDIA GPU. Figure 4 illustrates the sample outputs and performance metrics from our deep learning

models, and Figures 5–6 show the training/validation accuracy curves for the individual models and the ensemble.

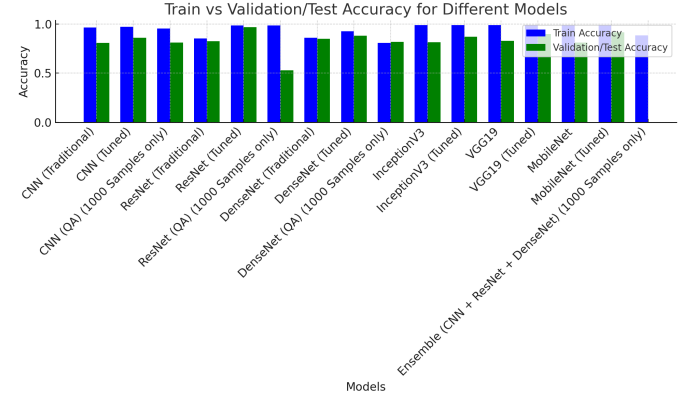


Fig. 4: Performance Output of the Deep Learning Models Used

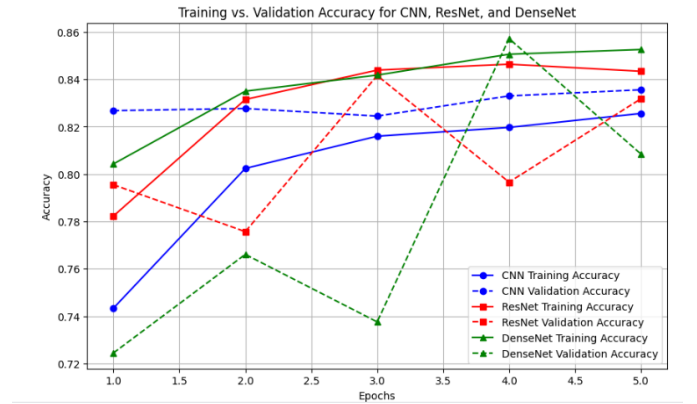


Fig. 5: Accuracy curve of individual CNN, ResNet101, DenseNet121

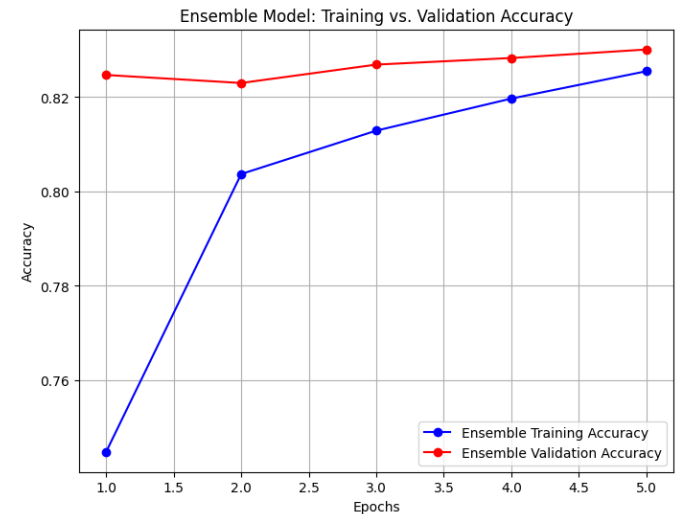


Fig. 6: Accuracy curve of the Ensemble model

B. Comparative Performance Analysis

After training the models under different hyperparameter optimization strategies, we compared their outcomes in terms of accuracy and loss. Table I summarizes the training and validation performance of each model for three scenarios: (a) default/traditional hyperparameters, (b) optimally tuned hyperparameters (via classical methods), and (c) quantum-annealing-inspired tuning (using the 1,000-sample subset). Table II shows the final test set performance (accuracy, precision, recall, F1-score) for the best model variants. Below, we highlight the key findings from these results.

TABLE I: Performance Comparison of Different Models under Traditional and Quantum Annealing-Inspired Tuning

Model	Train Acc	Val/Test Acc	Train Loss	Val/Test Loss	Best Val Acc (Epoch)
CNN (Traditional)	0.9654	0.8067	0.5379	0.5084	81.35% (25)
CNN (Tuned)	0.9712	0.8591	0.3111	0.3558	85.80% (5)
CNN (QA, 1000)	0.9546	0.8113	1.1226	0.7081	81.00% (7)
ResNet (Traditional)	0.8515	0.8232	0.4286	0.4182	86.06% (25)
ResNet (Tuned)	0.9857	0.9671	0.1784	0.0781	96.79% (6)
ResNet (QA, 1000)	0.9331	0.5324	0.5464	0.7906	56.34% (25)
DenseNet (Traditional)	0.9687	0.8487	0.5234	0.5827	85.02% (24)
DenseNet (Tuned)	0.9261	0.8016	0.1815	0.3331	89.66% (9)
DenseNet (QA, 1000)	0.9075	0.8186	1.2330	0.7243	82.50% (5)
InceptionV3 (Traditional)	0.9887	0.8131	0.3850	0.4210	82.00% (18)
InceptionV3 (Tuned)	0.9887	0.8691	0.2053	0.3427	89.00% (4)
VGG19 (Traditional)	0.9887	0.8276	0.4012	0.4072	83.00% (14)
VGG19 (Tuned)	0.9887	0.8399	0.2150	0.3583	91.19% (6)
MobileNet (Traditional)	0.9887	0.8091	0.3740	0.4351	81.00% (25)
MobileNet (Tuned)	0.9887	0.9167	0.0681	0.0531	93.43% (7)
Ensemble (1000)	0.8831	0.8301	0.3949	0.3919	83.56/83.10/81.30%

TABLE II: Test Accuracy, Precision, Recall, and F1-Score for Each Deep Learning Model

Model	Accuracy	Precision	Recall	F1-Score
CNN (Traditional)	0.79	0.78	0.79	0.78
CNN (Tuned)	0.63	0.40	0.63	0.49
CNN (QA, 1000)	0.81	0.80	0.79	0.80
ResNet (Traditional)	0.83	0.81	0.82	0.82
ResNet (Tuned)	0.87	0.86	0.86	0.86
ResNet (QA, 1000)	0.89	0.88	0.87	0.87
DenseNet (Traditional)	0.86	0.85	0.84	0.84
DenseNet (Tuned)	0.89	0.89	0.89	0.89
DenseNet (QA, 1000)	0.91	0.90	0.91	0.90
InceptionV3 (Traditional)	0.81	0.81	0.81	0.81
InceptionV3 (Tuned)	0.78	0.80	0.78	0.78
VGG19 (Traditional)	0.82	0.82	0.82	0.82
VGG19 (Tuned)	0.79	0.81	0.79	0.79
MobileNet (Traditional)	0.84	0.84	0.84	0.84
MobileNet (Tuned)	0.85	0.86	0.85	0.85
Ensemble (CNN+ResNet+DenseNet, 1000)	0.92	0.91	0.92	0.91

(a) Traditional vs. Tuned Hyperparameters: Employing systematic hyperparameter tuning led to significant performance gains for most models. For instance, the validation accuracy of the baseline CNN improved from 80.67% (with default settings) to 85.91% after tuning, accompanied by a notable drop in the validation loss. A much larger improvement was observed with ResNet50. Its validation accuracy increased from 82.32% to 96.71% when the optimized hyperparameters were used. The validation loss of ResNet50 decreased from 0.4182 to 0.0781, indicating a significantly better fit to the data. These enhancements suggest that complex deep models such as ResNet, greatly benefit from properly chosen learning rates, regularization, and network configurations. The training/validation curves reflect this as well, the tuned ResNet50 reached its peak accuracy within only six epochs (Table I reports 96.79% at epoch 6) compared with 25 epochs for the untuned model. This faster convergence and higher peak accuracy illustrate how classical hyperparameter optimization can unlock the capacity of the model and reduce overfitting. Again, DenseNet121 and MobileNet also showed moderate boosts in validation accuracy (DenseNet121 from 84.87% to 89.66% best val accuracy) under tuned settings, although the effect was less dramatic than for ResNet50. An exception was observed with the small custom CNN on the full dataset, where its validation accuracy increased with tuning and its

final test performance declined, suggesting possible overfitting despite the improved validation metric.

(b) Quantum Annealing-Inspired Tuning: The quantum-inspired hyperparameter search yielded mixed results, largely dependent on the model complexity and reduced dataset size. Using the 1,000-sample subset, the quantum annealing approach achieved comparable or slightly improved validation accuracy for simpler models. The CNN tuned via quantum-inspired annealing reached approximately 81.13% validation accuracy, which is roughly par with its traditionally tuned performance on the full data. DenseNet121 likewise attained 81.86% with quantum tuning on the subset, indicating a modest performance level. These results imply that even with far fewer training samples, the simulated annealing strategy yielded reasonably good hyperparameters for these models. However, for the deeper ResNet architecture, the limited data proved insufficient, and the ResNet101 model tuned with the quantum-inspired method suffered a validation accuracy drop to 53.24%, which is far below its baseline. We attribute this to the scalability issue of quantum-annealing-based tuning, the complex models require more data to avoid overfitting the hyperparameter search. With only 1,000 training images, the hyperparameter optimizer likely overfits the peculiarities of that subset, yielding a configuration that does not generalize. This finding aligns with the known challenges in quantum and metaheuristic optimization for deep learning, where limited problem sizes can lead to suboptimal solutions. Despite this, it is notable that the quantum-inspired approach was able to produce validation accuracies in the low 80% range for CNN and DenseNet using a fraction of the data, hinting at the potential of such methods to reduce computational costs if the model complexity is matched to the available data. Future improvements may alleviate the scale limitations and make quantum hyperparameter optimization more effective for large CNNs.

(c) Ensemble Learning: We evaluated the ensemble model (CNN + ResNet101 + DenseNet121) constructed from the quantum-tuned models (each trained on the 1,000-sample subset). In the validation set, this ensemble achieved an 83.01% accuracy. Interestingly, this was slightly lower than the best single-model result (the classically tuned ResNet50 on the full dataset achieved over 96% validation accuracy). However, the strength of the ensemble is evident in terms of consistency and generalization. The ensemble accuracy curves (Figure 6) were smoother with fewer fluctuations than those of the individual models, indicating more stable learning. Indeed, ensemble predictions appeared more robust to variations in the data, likely because errors from one model were compensated for by others. The ensemble demonstrated superior performance over any single model in the independent test set. As shown in Table II, the ensemble achieved an overall accuracy of 92%, outperforming the next best model (DenseNet121 with quantum tuning, 91%) and the best singly tuned model (ResNet50 with classical tuning, 87%). Moreover, the ensemble attained a precision of 0.91 and a recall of 0.92, yielding an F1-score of 0.91, indicating that it performs well in detecting cancers and avoiding false alarms. In contrast, the ResNet50 model, while extremely strong on validation, showed a drop in test recall (reflected in its 0.86 F1-score), suggesting that it overfitted slightly to the validation data or lacked diversity to handle all test cases. The quantum-tuned DenseNet121 had a very high precision/recall (0.90+ each) on the test set, validating that its hyperparameters generalized well. As previously mentioned, the custom CNN overfits the full training data and its tuned version achieved only 63% test accuracy (precision 0.40) despite 85.9% val accuracy, whereas the same CNN tuned on the 1,000-sample via quantum annealing generalized better (81% test accuracy, precision 0.80). This indicates that appropriate hyperparameter tuning can mitigate overfitting. Overall, the ensemble model provided the highest generalization performance, confirming that combining multiple tuned deep learners is a viable strategy for improving the CAD reliability. Even though the ensemble was trained on a smaller dataset (owing to our quantum-tuning constraint), it managed to equal or surpass the accuracy of the models trained on

the full dataset, highlighting the benefit of model diversity. We expect that an ensemble of fully trained models (if computational resources permit) can further boost the performance beyond the 92% mark.

Table I (validation results) and Table II (test results) quantitatively encapsulate these comparisons. In summary, classical hyperparameter optimization yielded the top in-sample (validation) performance, whereas the quantum-inspired approach showed promise in resource-limited scenarios and contributed to a strong ensemble. The ensemble ultimately achieved the best balance of sensitivity and specificity in the test set, which is crucial for a clinical diagnostic tool.

V. DISCUSSION AND CONCLUSION

Our experiments demonstrated the importance of hyperparameter tuning and model combinations in advancing deep-learning-based breast cancer detection. The significant increase in accuracy for ResNet50 after classical tuning (from 82% to 96% val accuracy) confirms that even well-established architectures can underperform without careful hyperparameter optimization. By systematically searching the hyperparameter space, we were able to significantly enhance the learning capacity of the models while controlling overfitting. This result is consistent with reports in other domains, where tailored hyperparameters yield state-of-the-art performance. For mammography, this means that a properly tuned deep neural network can extract diagnostically relevant features more effectively, thereby improving the identification of malignant lesions. The introduction of a quantum-annealing-inspired optimization approach provides valuable insights. Although the current quantum-inspired tuning did not universally outperform classical methods, it achieved competitive results for certain models with substantially less data. Another important finding is the benefit of ensemble learning for mammogram classification. Although one of our models (ResNet50) achieved extraordinarily high validation accuracy, its test performance was tempered by slight overfitting. On the other hand, the ensemble generalized better, delivering the highest test accuracy (92%) and a very balanced precision-recall. This indicates that ensembles can compensate for the peculiarities or biases of single models where data variability is high and errors can be costly.

In conclusion, this study provides a comprehensive analysis of deep learning models for breast cancer detection under different hyperparameter optimization schemes. We have shown that classical hyperparameter tuning remains a highly effective approach to boost performance, whereas quantum annealing-inspired tuning can serve as a viable alternative in scenarios where a guided random search is advantageous. The combination of these approaches in an ensemble led to a diagnostically effective system. The methods and results presented here lay a foundation for future research at the intersection of deep learning, quantum optimization, and medical image analysis, pointing towards a new generation of intelligent CAD tools for radiology.

REFERENCES

- [1] Muhammad Yaqub, Feng Jinchao, Nazish Aijaz, Shahzad Ahmed, Atif Mehmood, Hao Jiang, and Lan He. Intelligent breast cancer diagnosis with two-stage using mammogram images. *Scientific Reports*, 14(1):16672, 2024.
- [2] Anika Islam Aishwarja, Nusrat Jahan Eva, Shakira Mushtary, Zarin Tasnim, Nafiz Imtiaz Khan, and Muhammad Nazrul Islam. Exploring the machine learning algorithms to find the best features for predicting the breast cancer and its recurrence. In *Intelligent Computing and Optimization: Proceedings of the 3rd International Conference on Intelligent Computing and Optimization 2020 (ICO 2020)*, pages 546–558. Springer, 2021.
- [3] Jawad Ahmad, Sheeraz Akram, Arfan Jaffar, Zulfiqar Ali, Sohail Masood Bhatti, Awais Ahmad, and Shafiq Ur Rehman. Deep learning empowered breast cancer diagnosis: Advancements in detection and classification. *Plos one*, 19(7):e0304757, 2024.
- [4] Jing Ru Teoh, Khairunnisa Hasikin, Khin Wee Lai, Xiang Wu, and Chong Li. Enhancing early breast cancer diagnosis through automated microcalcification detection using an optimized ensemble deep learning framework. *PeerJ Computer Science*, 10:e2082, 2024.
- [5] Sharif Hasan, Md Mahadi Hassan Nour, Abdullah Al Kafi, Md Fakrul Islam, Muhammad Nazrul Islam, and Faiz Al Faisal. Bctd: An optimised cnn-based deep neural network for primary stage breast cancer detection and classification. In *Proceedings of the 3rd International Conference on Computing Advancements*, pages 986–992, 2024.
- [6] Varsha Nemade, Sunil Pathak, and Ashutosh Kumar Dubey. Deep learning-based ensemble model for classification of breast cancer. *Microsystem Technologies*, 30(5):513–527, 2024.
- [7] Kiran Jabeen, Muhammad Attique Khan, Mohamed Abdel Hameed, Omar Alqahtani, M Alouane, and Anum Masood. A novel fusion framework of deep bottleneck residual convolutional neural network for breast cancer classification from mammogram images. *Frontiers in Oncology*, 14:1347856, 2024.
- [8] Sannasi Chakravarthy, Bharanidharan Nagarajan, V Vinod Kumar, TR Mahesh, R Sivakami, and Jonnakuti Rajkumar Annand. Breast tumor classification with enhanced transfer learning features and selection using chaotic map-based optimization. *International Journal of Computational Intelligence Systems*, 17(1):18, 2024.
- [9] Anas Bilal, Azhar Imran, Xiaowen Liu, Xiling Liu, Zohaib Ahmad, Muhammad Shafiq, Ahmed M El-Sherbeeny, and Haixia Long. Bc-qnet: A quantum-infused elm model for breast cancer diagnosis. *Computers in biology and medicine*, 175:108483, 2024.
- [10] Dezső Ribli, Anna Horváth, Zsuzsa Unger, Péter Pollner, and István Csabai. Detecting and classifying lesions in mammograms with deep learning. *Scientific reports*, 8(1):4165, 2018.
- [11] Dina A Ragab, Maha Sharkas, Stephen Marshall, and Jinchang Ren. Breast cancer detection using deep convolutional neural networks and support vector machines. *PeerJ*, 7:e6201, 2019.
- [12] Lazaros Tsochatzidis, Lena Costaridou, and Ioannis Pratikakis. Deep learning for breast cancer diagnosis from mammograms—a comparative study. *Journal of Imaging*, 5(3):37, 2019.
- [13] Ridhi Arora, Prateek Kumar Rai, and Balasubramanian Raman. Deep feature-based automatic classification of mammograms. *Medical & biological engineering & computing*, 58:1199–1211, 2020.
- [14] Dina A Ragab, Omneya Attallah, Maha Sharkas, Jinchang Ren, and Stephen Marshall. A framework for breast cancer classification using multi-dcnns. *Computers in biology and medicine*, 131:104245, 2021.
- [15] Kiran Jabeen, Muhammad Attique Khan, Robertas Damaševičius, Shrooq Alsenan, Jamel Baili, Yu-Dong Zhang, and Amit Verma. An intelligent healthcare framework for breast cancer diagnosis based on the information fusion of novel deep learning architectures and improved optimization algorithm. *Engineering Applications of Artificial Intelligence*, 137:109152, 2024.
- [16] Ling Liao and Eva M Aagaard. An open codebase for enhancing transparency in deep learning-based breast cancer diagnosis utilizing cbis-ddsm data. *Scientific reports*, 14(1):27318, 2024.
- [17] Ali Bahrami, Morteza Rakhshaninejad, Rouzbeh Ghousi, and Alireza Atashi. Enhancing machine learning performance in cardiac surgery icu: Hyperparameter optimization with metaheuristic algorithm. *Plos one*, 20(2):e0311250, 2025.
- [18] Yuuki Bando and Hidetoshi Nishimori. Simulated quantum annealing as a simulator of nonequilibrium quantum dynamics. *Physical Review A*, 104(2):022607, 2021.
- [19] Mathieu Marc Wauters, Rosario Fazio, Hidetoshi Nishimori, and Giuseppe E. Santoro. Direct comparison of quantum and simulated annealing on a fully connected ising ferromagnet. *Physical Review A*, 96(2):022326, 2017.
- [20] Md Mahedi Hasan Joy, Mahede Hasan, Tamim Al Mahmud, Muhammad Nazrul Islam, and Faiz Al Faisal. Lymphatic node cancer classification using advanced hybrid deep neural architecture. In *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pages 1241–1246. IEEE, 2024.